Date Functions

Table 1: Employees

emp_id	name	hire date
1	Alice	2020-01-15
2	Bob	2021-06-10
3	Charlie	2023-03-22

Q1 Add 6 monts to each employees hire date using DATEADD
Expected columns: emp_id, name, hire_date hireplus 6 months

```
SELECT emp_id,
       name,
       hire_date,
    DATEADD(Month,6,hiredate) AS hire plus 6months
FROM Employees;
```

EmpId	name	hire date	hire plus 6 monts
1	Alice	2020-01-15	2020-07-15
2	Bob	2021-06-10	2021-12-10
3	Charlie	2023-03-22	2023-09-22

Q2 Ages in days from dob to today

```
SELECT student_id,
    name
    DATEDIFF(DAY,dob, CURRENT_DATE) AS ageindays
FROM Students;
```

Output (Assuming today = 2025-10-13)

Student_id	name	age in days
101	Maya	7375
102	Ethan	7669
103	Sienna	7169

Q3:
Events

Event_id	event_name	event date
1	Seminar	2024-06-15
2	Workshop	2025-07-10
3	Hackathon	2025-04-20

Find how many days are left until each event using DATEDIFF()

Event_id	event_name	day remaining
1	Seminar	-485
2	Workshop	-42
3	Hackathon	-266

```
SELECT event_id,
       event_name
       DATEDIFF(Day, Current_date, event_date) AS day_remaining
FROM Events;
```

Q4 Invoices

Invoice_id	Issue_date	due date
501	2025-03-10	2025-03-25
502	2025-04-01	2025-04-15
503	2025-04-10	2025-04-20

Calculate the number of days between issuedate and duedate

```
SELECT invoice_id,
       issue_date,
       due date,
       DATEDIFF(Day, issuedate, due date)
FROM   invoices
```

Invoice_id	Issuedate	due date	days_between
501	2025-03-10	2025-03-25	15
502	2025-04-01	2025-04-15	14
503	2025-04-10	2025-04-20	10

Q5 Courses

Course_id	name	startdate
201	SQl Basics	2025-05-01
202	Python	2025-06-01

Format start date as Month YYYY using TO_CHAR()

```
SELECT course id,
       name,
       TO CHAR( Start_date 'Month YYYY')
FROM   courses;
```

Course_id	name	Formatteddate
201	SQL Basics	May 2025
202	Python	June 2025

Q6 Memberships

Member_id	Start year	start month	start_day
1	2023	5	10
2	2022	11	25

Create full date from parts using DATE FROMPARTS

```
SELECT member_id
       DATEFROMPARTS(Start_year ,start month, start day)
       AS   full start date
FROM   Memberships;
```

Member_id	Full start_date
1	2023-05-10
2	2022-11-25

Q7: Subscriptions

Sub-id	Plan	Renewal date
11	Basic	2025-01-01
12	Premium	2025-03-15

Extend each renewal date by 1 year using DATEADD()

```
SELECT sub-id,
       Plan,
       DATEADD (Year, 1, renewal-date) AS Extended renewal_date
FROM Subscptions;
```

Sub-id	Plan	Extended renewal_date
11	Basic	2026-01-01
12	Premium	2026-03-15

Q8 Orders

Order_id	order_date
1001	2025-04-15
1002	2025-04-10

Show current date & difference from order_date

Use current_date and DATEDIFF()

```
SELECT order_id,
       Order_date,
       CurrENT DATE AS Today date,
DATEDIFF(DAY, order_date, CURRENT DATE) AS day sia
order
FROM order;
```

order_id	order date	today date	Days since order
1001	2025-04-15	2025-10-13	181
1002	2025-04-10	2025-10-13	186

Q9

```
SELECT training_id,
       topic
       EXTRACT (YEAR FROM training date) As training_year
FROM Trainings;
```

Training-id	Topic	Training year
1	Safety	2025
2	Compliance	2025

Q10

```
SELECT post_id,
       title,
       EXTRACT( HOUR FROM published on As hour published
       EXTRACT( MINUTE from published on ) As minute
       published
FROM Blog-Posts;
```

Post_id	title	hour published	minutes-published
1	SQL Tips	10	15
2	Data cleaning	16	45

Q11

```
SELECT driver_id,
       license_expiry,
       DATEDIFF (DAY, CURRENT DATE, license_expiry) AS day_left
FROM Drivers;
```

Driver_id	License expiry	Day left
301	2025-08-10	-64
302	2023-12-31	-652

Q12

```
SELECT message-id,
       sent_timestamp,
       CURRENT TIMESTAMP AS current timestamp,
       DATEDIFF (Second, sent_timestamp, CURRENT TIMESTAMP)
       AS seconds_since sent
FROM Messages;
```

Message_id	Sent_timestamp	Current timestamp
1	2025-04-19 09:32:45	2025-10-13 13:00:00
2	2025-04-18 23:59:59	2025-10-13 13:00:00

Q13

```
SELECT return_id,
       return_date,
       DATEADD( DAY, 15, return date) AS restock_date
FROM Returns;
```

return_id	return_date	Restock date
901	2025-04-05	2025-04-20
902	2025-04-01	2025-04-16

Q14

```
SELECT Assign_id,
       TO DATE(Assigned on, YYYY-MM-DD') AS
       assigned on date
FROM Assgnments;
```

Assign id	Assigned on_date
1	2025-03-01
2	2025-03-05

Q15

```
SELECT meeting_id,
       TO CHAR(scheduled_time, Month DD, YYYY at
            HH12:MI AM') AS formatted_meeting_time
FROM meetings;
```

meeting id	Formatted meeting time
1	April 19, 2025 at 02:00 PM
2	April 19, 2025 at 09:30 AM

Exercise 2.1

1 Show all employees with (0 if NULL)

```
SELECT employee_id,
       name,
       IFNULL(Salary, 0) AS Salary with default
  FROM employees;
```

Employee-id	Name	Salary with default
1	Alice	5000
2	Bob	0
3	Charlie	7000
4	Dana	0

2 Department name or "Not Assigned"

```
SELECT employee_id,
       name,
       IFNULL(Department 'Not Assigned') AS Departname
FROM employees;
```

1	Alice	HR
2	Bob	IT
3	Charlie	Not Assigned
4	Dana	Finance

3. Orders with NULL customer id

```
SELECT order_id, customer_id
  FROM Orders
 WHERE customer_id IS NULL;
```

Order id	Customer_id
103	NULL

4. Delivery status (Pending if NULL)

```
SELECT order_id
       Customer_id
   IFNULL(TO CHAR(delivery_date, YYYY-MM-DD'), Pending
  AS delivery status
FROM Orders;
```

Order id	Customer_id	Delivery status
101	201	2024-12-01
102	202	Pending
103	NULL	2024-12-03

5. Replace NULL grades with 0

```
SELECT student id,
       name,
       IFNULL(Grade, 0) AS final grade
FROM Students;
```

Student_id	name	final_grade
1	Ethan	85
2	Maya	0
3	Olivia	90

6. Count students with NULL grade

```
SELECT count(*) AS not graded count
FROM students
WHERE grade is NULL
```

Not graded count
1

Table 4: Products

7. Final price after discount

```
SELECT product_id,
       name,
  Price - IFNULL (discount, 0) AS Final price
FROM Products;
```

Product_id	name	final price
501	Keyboard	25
502	Mouse	10
503	Moniter	100

Table 5

8. Count customers with no email

```
SELECT count (*) AS missing email_count
FROM Customers
WHERE email IS NULL;
```

Missing email_count
2

9. Show all customers with email or No email

```
SELECT customer id,
       name
   IF NULL (email, 'No email') AS email display
FROM Customers;
```

Customer-id	name	email display
1	Linda	No emeal
2	Joseph	Josph@emal.com
3	Nia	No email

Table 6

10. Replace NULL method with Unknown

```
SELECT payment_id,
  IFNULL(Method 'Unknown') AS Method display,
  Status
FROM payments;
```

Payment_id	Method display	Status
301	Credit	NULL
302	PayPal	Suscess
303	Unknown	Failed

Table 7: Inventory

Item_id	Item name	Quantity
1	Pen	NULL
2	Notebook	150
3	Eraser	NULL

```
SELECT item id,
       item_name
  IFNUL(Quantity, 0) AS quantity checked
FROM Inventory;
```

1	Pen	0
2	Notebook	150
3	Eraser	0

Table8 Employee_Extra

12.

OR commission

```
SELECT emp_id,
       COALSCE(bonus, commission, 0) AS first available rewa
FROM Employees_Extra;
```

Emp_id	First available reward
1	300
2	100
3	0

Table 9 : Classes

Q13

```
SELECT count(*) AS no room count
FROM Classes
WHERE room is NULL;
```

no_room count
2

Table 10: Attendance

Q14

```
SELECT Student_id,
       date,
       IFNULL (status 'Not Marked') AS Attendan
       status
FROM Attendance
```

Studentid	date	Attendance_status
1	2025-04-01	Not marked
2	2025-04-01	Present
3	2025-04-01	Absent

Table 11 : Bank Accounts

Q15

```
SELECT account_id,
    IFNULL (Account type 'Unknown') As type display,
    IFNULL (Balance, 0) As balance_checked
FROM Bank accounts;
```

Account_id	type display	balance checked
A1	Savings	0
A2	Current	5000
A3	Unknown	2000

Table 12 Projects

Q16

```
SELECT project_id,
        title,
    IFNULL (TO CHAR (Start_date 'YYYY-MM-DD') TBD
  As start display
FROM projects;
```

Projects_id	title	Start display
1	Website revamp	2025-01-10
2	Mobile App	TBD
3	Data Migration	TBD

Q17

```
SELECT review_id,
       Product_id,
    IFNULL (Comment, 'No comment') AS comment_display,
   IFNULL (rating, 0) AS rating_display
  FROM Reviews;
```

Review_id	Product_id	Comment_display	rating_display
1	501	Great product	4
2	502	No comment	0
3	503	Work fine	3

Q18

```
SELECT supplier_id,
       name,
    COALSEE (Phone, Alt-phone, No contact) AS contact numb
FROM   Suppliers;
```

Suppler_id	Name	Contact number
1	Global goods	123456789
2	Best supplies	98765421
3	Valuesourse	No contact

Q19

```
SELECT  user_id,
   IF NULL (theme 'Light') AS timeset,
   IF NULL ( Language 'English) As Language set,
   IF NULL (timezone UTC As timezone set
FROM  User settings;
```

Q17

```
SELECT review_id,
       Product_id,
   IF NULL (Comment, 'No comment') AS comment_display,
  IFNULL (rating, 0) AS rating display
FROM Reviews;
```

Review_id	Product_id	Comment display	rating_display
1	501	Great product	4
2	502	No comment	0
3	503	Work fine	3

Q18

```
SELECT supplier_id,
       name,
   COALSEE (Phone, Alt-phone, No contact) AS contact number
FROM Suppliers;
```

Suppler_id	Name	Contact number
1	Global goods	123456789
2	Best supplies	98765421
3	Valuesarse	No contact

Q19

```
SELECT user_id,
  IF NULL (theme 'Light') AS timeset,
  IF NULL ( Language 'English) AS Language set,
  IF NULL (timezone UTC AS timezone_set
FROM User settings;
```

User_id	Theme set	Language set	time zone
1	Light	English	UTC
2	Dark	English	UTC +1
3	Light	English	UTC

Q20

```
SELECT record_id,
       machine_id,
   IFNULL( issue, unknown Issue) AS issue log,
   IFNULL ( technical 'Not Assigned ) AS Technican_name
FROM  Maintenance;
```

record id	Machine id	issue log	Technical name
1	M101	Overheating	Not assigned
2	M102	Unknown issue	Not assigned
3	M103	Jammed	Alex